Exercice 3 :

a)

$$Y_t - 0,5\,Y_{t-1} - Y_{t-4} + 0,5\,Y_{t-5} = e_t - 0,3\,e_{t-1}$$

$$(1 - 0,5B - B^4 + 0,5B^5)\,Y_t = (1 - 0,3B)\,e_t$$

$$(1 - 0,5B)(1 - B^4)\,Y_t = (1 - 0,3)\,e_t$$

So Y_t is an $ARIMA(1,0,0) \times (0,0,1)_4$

$$Y_t = Y_{t-1} + Y_{t-12} - Y_{t-13} + e_t - 0,5\,e_{t-1} - 0,5\,e_{t-12} + 0,25\,e_{t-13}$$

$$(1 - B - B^{12} + B^{13})\,Y_t = (1 - 0,5B - 0,5B^{12} + 0,25B^{13})\,e_t$$

$$(1 - B)(1 - B^{12})\,Y_t = (1 - 0,5B)(1 - 0,5B^{12})\,e_t$$

So $d = 1$, $s = 12$, $q = 1$, $Q = 1$ and $p = 0$

So Y_t is an $ARIMA(0,1,1) \times (0,1,1)_{12}$